

23/40

MSc(IT) Semester II [Winter 2025-26]

163907

IT632: Software Engineering

18-Mar-2026

Duration: 90 minutes

Max. Marks: 40

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4 Use Pen for writing Answer !!

1. Multiple Choice Questions - Select the most appropriate choice.

(1x5 = 5 Marks)

1.1 In the **Basic COCOMO** model, if a project is classified as "**Embedded**" rather than "Organic," how does this classification specifically affect effort estimation?

- A) It decreases the constant a because embedded systems are more efficient to develop.
- ☒ B) It increases both the constant a (to 3.6) and the exponent b (to 1.20) to reflect higher complexity and more rigid constraints.
- C) It only changes the input from KLOC to Function Points.
- D) It requires the use of 15 cost drivers instead of the basic formula.

1.2 When calculating the final **Function Points (FP)**, how is the **Technical Complexity Factor (TCF)** utilized in relation to the **Unadjusted Function Points (UFP)**?

- A) The TCF is added to the UFP to account for environmental buffers.
- B) The UFP is divided by the TCF to normalize the size across different programming languages.
- ☒ C) The FP is calculated by multiplying the UFP by the TCF, which is derived from 14 general system characteristics.
- D) The TCF is only used if the programming language is unknown.

1.3 A module named `SystemStartupModule` contains the functions `initializeMemory()`, `loadConfig()` and `displayWelcomeScreen()`. Assume that these tasks are conceptually unrelated. What type of **cohesion** does this represent?

- (A) Functional
- ☒ (B) Sequential
- (C) Procedural
- (D) Temporal

1.4 If a `BillingModule` passes a **complete "Customer" object** (containing name, address, ID, and history) to a `NotificationModule` that **only needs the "Email Address" field** to send a message, which type of coupling is being exhibited?

- (A) Data
- ☒ (B) Stamp
- (C) Common
- (D) Control

1.5 Why is **Function Point Analysis (FPA)** often used as a precursor to **COCOMO** for effort estimation in the **early stages** of the software development lifecycle?

- A) Because COCOMO is only used for time estimation, while FPA is for effort.
- B) Because FPA automatically calculates the cost drivers (EAF) required for Intermediate COCOMO.
- ☒ C) Because FPA provides a language-independent measure of functional size that can be used to estimate Lines of Code (LOC), which is the required input for COCOMO.
- D) Because FPA is subjective and COCOMO is objective, so they must be averaged.

Software Specification: Skill-Based Gaming & Betting Platform (SBGBP)

SBGBP is a mobile-first, real-money skill-gaming platform that enables registered users to participate in competitive games, fantasy sports contests, and head-to-head betting challenges across multiple game categories including cricket fantasy, card games (Rummy, Poker), puzzle challenges, and trivia tournaments. The system supports user registration with KYC (Know Your Customer) verification, wallet management with secure deposit and withdrawal via UPI, net banking, and cards through integrated payment gateways, and real-time contest matchmaking based on entry fee, skill tier, and game type.

Users can browse active contests, join or create private rooms, and track live leaderboards during gameplay. The platform manages prize pool calculation, tax deduction (TDS), and automated winnings disbursement upon contest completion. A fraud detection and anti-cheat module monitors gameplay patterns, flags anomalies, and enforces platform fairness policies. Referral and loyalty reward systems incentivize engagement. Administrators oversee user management, contest configuration, financial reconciliation, and compliance reporting. Post-contest processes include result verification, rating updates, and feedback collection.

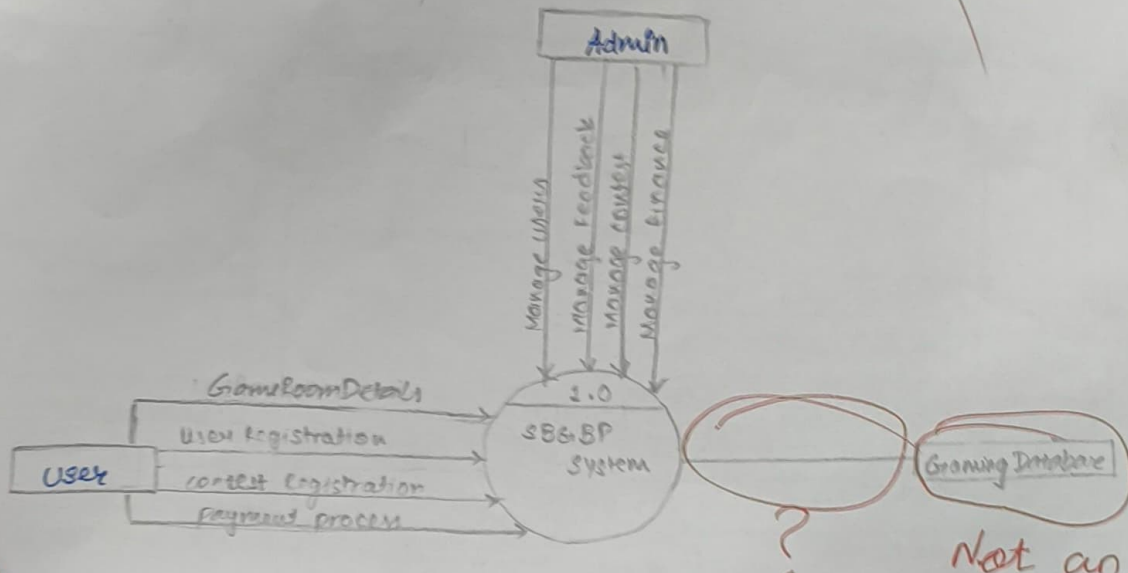
Attempt Q.2 to Q.7 based on the above given SBGBP specification.

External Entities	Processes	Data Stores
User	User Registration	User Details
Admin	Payment Process	Payment Details
	Contest Registration	Contest Details
		Feedback Details

User Reg
☒ KYC
☒ wallet management
☒ real-time monitoring

2. Design Context-level diagram for SBGBP.

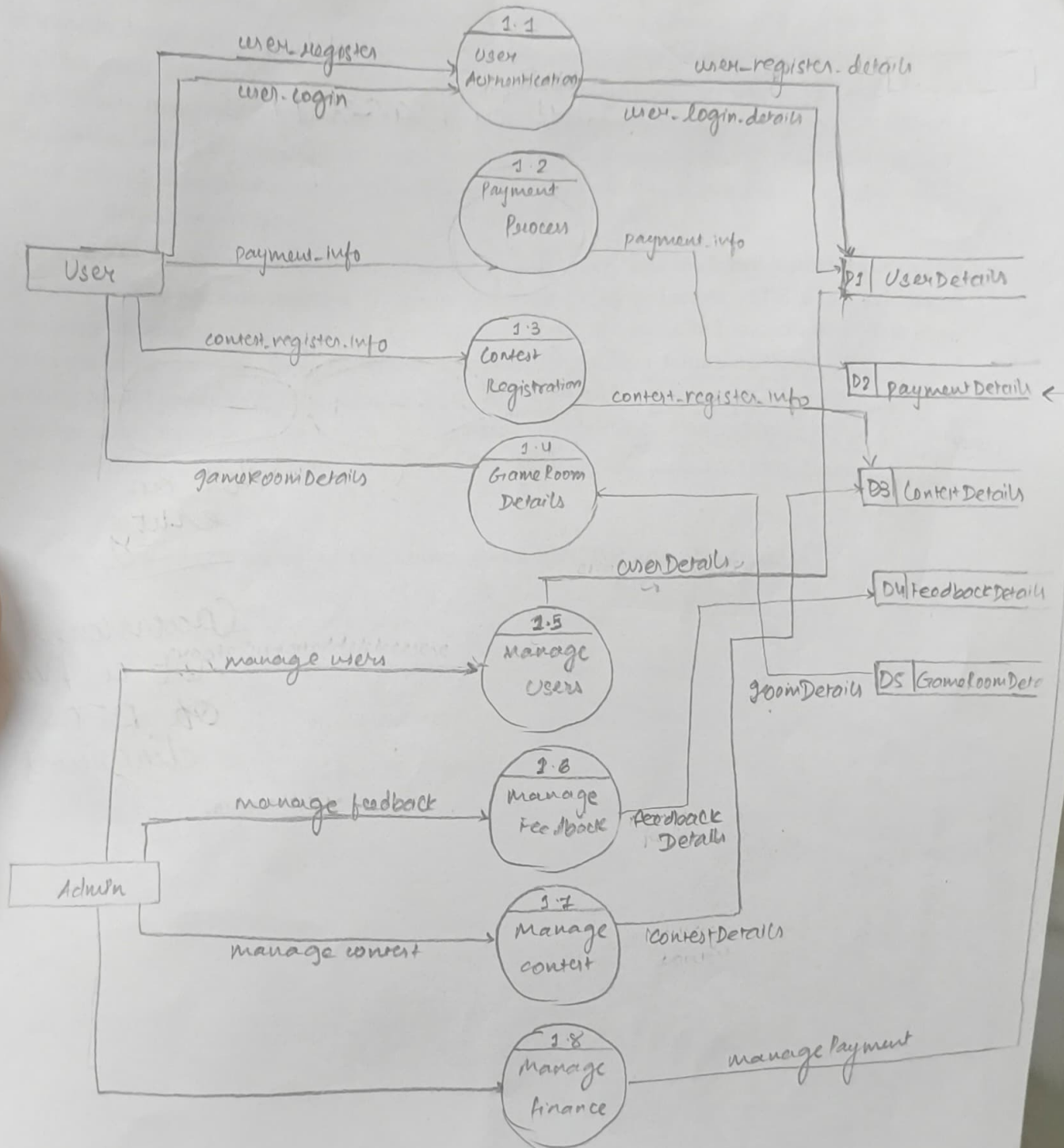
(4 Marks)



Not an
entity

(Data stores
not a part
of DFD-D
diagrams)

3. Based on identified key processes (atleast 8), and datastores (atleast 5) construct DFD Level 1 model for SBGBP. (5 Mark)

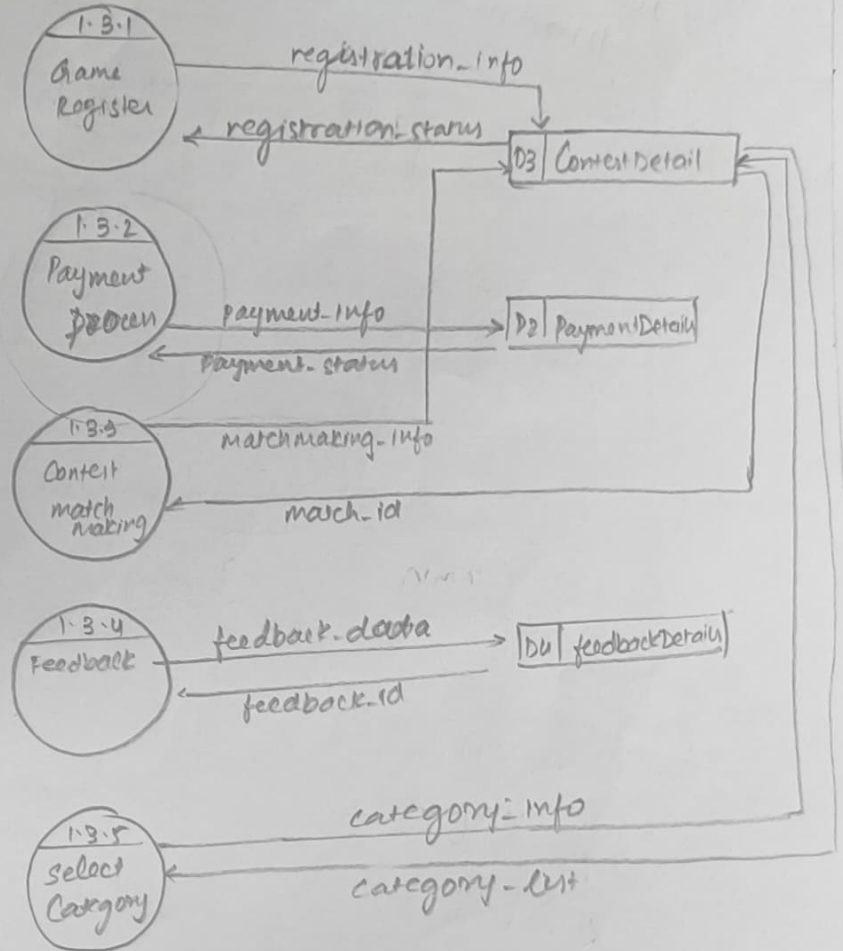


41. Design DFD Level 2 for contest lifecycle management in SBGBP.

2

(4 Marks)

user ??

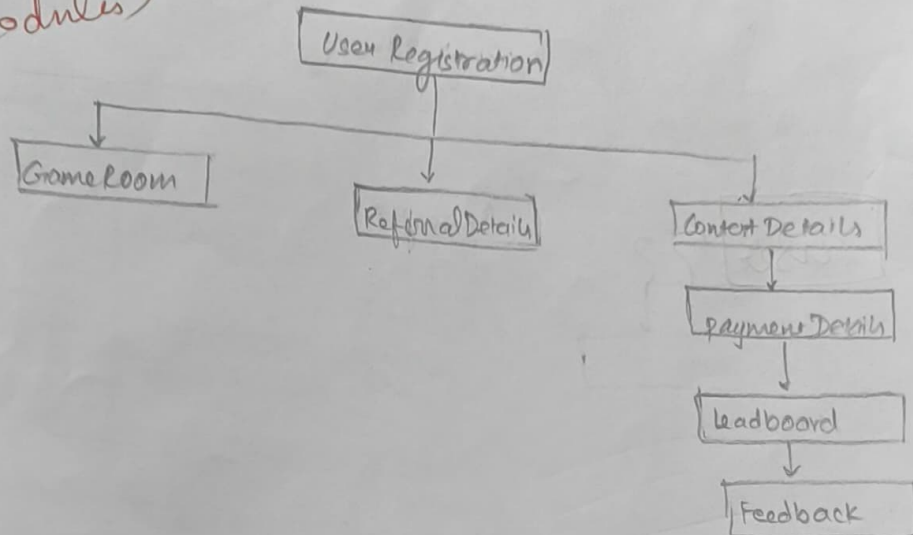


- Include entities

5. Design module structure chart for SBGBP aligning your DFD Level 1.

(4 Marks)

2
(no sub modules)

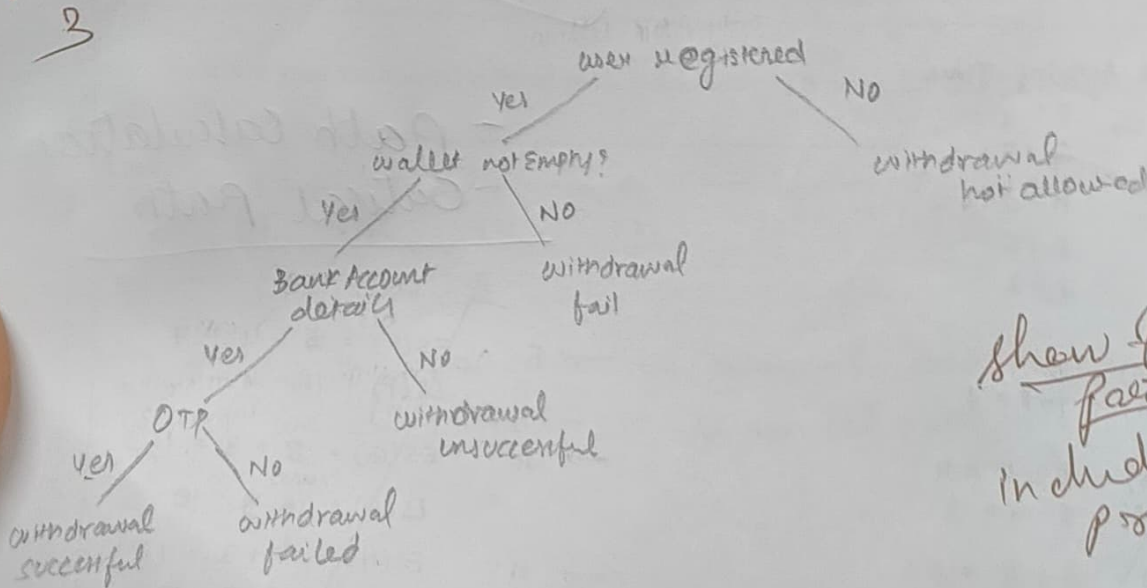


6. Construct decision table for the KYC Approval process in SBGBP. All boxes for conditions and rules may not necessarily be filled. (4 Marks)

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Registered	✓	✓	✓	X
Aadhar Card No.	✓	X	✓	X
OTP	✓	X	X	X
process successful Action	✓	X	X	X

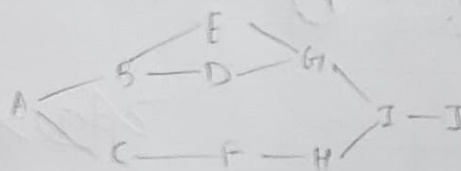
Age 18
Residence
Proof

7. Construct decision tree for the Withdrawal Authorization process in SBGBP. (4 Marks)



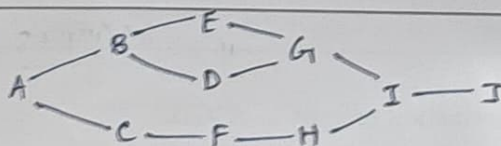
show grand
parent.
include make
process in
SBGBP.

8. For the following activities, compute expected activity time, possible project paths, path durations, and critical path. Give interpretation for your computations. (5 Mark)



Activity	Predecessor	O	M	P	EAT	Slack
A	-	2	4	6	4	2
B	A	2	3	5	3-16	-
C	A	3	4	7	4-33	-
D	B	2	4	8	4-33	-
E	B	3	5	9	5-33	-
F	C	2	4	6	4	-
G	D,E	2	3	5	3-16	-
H	F	2	3	4	3	-
I	G,H	3	5	9	5-33	-
J	I	1	2	3	2	-

Ans:



possible paths

~~A-B-E-G-I-J~~
~~A-B-D-G-I-J~~
~~A-C-F-H-I-J~~

Expected Activity Time

A : 4 F : 4
 B : 3 G : 3
 C : 4 H : 3
 D : 4 I : 5
 E : 5 J : 2

- Path Calculation
 - Critical Path

→ A: $ES(A) = 0 - 0 = 0$
 $LS(A) = 5 - 3 = 2$

→ B: $ES(B) = 0 + 4 = 4$
 $LS(B) = 9 - 4 = 5$

→ C: $ES(C) = 0 + 3 = 3$
 $LS(C) = 10 - 3 = 7$

→ D: $ES(D) = 4 + 4 = 8$
 $LS(D) = 15 - 4 = 11$

→ E: $ES(E) = 4 + 5 = 9$
 $LS(E) = 19 - 5 = 14$

→ F: $ES(F) = 3 + 4 = 7$
 $LS(F) = 10 - 4 = 6$

→ G: $ES(G) = 8 + 3 = 11$
 $LS(G) = 15 - 3 = 12$

→ H: $ES(H) = 7 + 3 = 10$
 $LS(H) = 13 - 3 = 10$

→ I: $ES(I) = 11 + 5 = 16$
 $LS(I) = 22 - 5 = 17$

→ J: $ES(J) = 16 + 2 = 18$
 $LS(J) = 18$

Ans:

FS

$$\begin{aligned}
 A &: 0 + 4 = 4 \\
 B &: 4 + 3 = 7 \\
 C &: 4 + 4 = 8 \\
 D &: 7 + 4 = 11 \\
 E &: 7 + 5 = 12 \\
 F &: 8 + 4 = 12 \\
 G &: 12 + 3 = 15 \\
 H &: 12 + 3 = 15 \\
 I &: 15 + 5 = 20 \\
 J &: 20 + 2 = 22
 \end{aligned}$$

LS

$$\begin{aligned}
 A &: 9 - 4 = 5 \\
 B &: 12 - 3 = 9 \\
 C &: 13 - 4 = 9 \\
 D &: 17 - 4 = 13 \\
 E &: 17 - 5 = 12 \\
 F &: 17 - 4 = 13 \\
 G &: 20 - 3 = 17 \\
 H &: 20 - 3 = 17 \\
 I &: 22 - 2 = 20 \\
 J &: 22
 \end{aligned}$$

Slack

$$\begin{aligned}
 A &: 1 \\
 B &: 2 \\
 C &: 1 \\
 D &: 2 \\
 E &: 0 \\
 F &: 1 \\
 G &: 2 \\
 H &: 2 \\
 I &: 0 \\
 J &: 0
 \end{aligned}$$

extra !!

9. Verify/Validate your computed critical path from above Q.8 using ES and LS computations. (5 Marks)

Ans:

$$A \xrightarrow{(1)} B \xrightarrow{(2)} E \xrightarrow{(0)} G \xrightarrow{(2)} I \xrightarrow{(0)} J \xrightarrow{(0)} \text{Critical path}$$

Slack time: 5

→ This is the path with the minimal slack time of 5 days so we can consider it as critical path.